The Order of the Silver Padlock

Instructor print pack · single-use pages · print once, double-sided

How to print. Print the whole pack **double-sided, flip on the long edge**, once. Guide, key and list pages share sheets with other guide pages, so they may sit next to a different activity's page. **Anything you cut apart, tape up or hand out has a blank back**, so cutting a card never destroys content on the other side. Print cards and signs on card stock if you can. The Warden worksheets are in the separate **Per-Warden Print Pack**.

Activity	Req	Pages	What to do
Line of Oaths	2b	2-3, 5, 7, 9	1 set. Tape the 3 signs to the wall; cut apart the scenario cards; keep the key.
Hunt for Hidden Doors	1b	11, 13, 15	1 copy for you (guide, call-out list, tally sheet).
Three Pillars Trial	3b	12, 17, 19, 21	1 set. Hang the C, I and A signs in three corners; keep the scenario key.
Locked Vault	4b	23, 25, 27, 29	1 set. Cut apart the ransom notes and decision cards (one set per lodge).
Scrying Glass Demo (Wireshark)	4c	24	1 copy for you.
Masked Messengers	4d	31, 33, 35, 37	1 set. Cut apart the PHISH and LEGIT cards (one pair per lodge).
Count the Open Gates	4f	32, 39, 41, 43	1 set. Key plus lodge cards. (The Warden sheet is in the per-Warden pack.)
Herald Dispatch	4e	45	Instructor guide only. The articles and answer key are not printed: share each lodge its article on a device (full file in Final Docs). The report sheet is in the per-Warden pack.
Passphrase Forge	5c	46-47, 49, 51	1 set. Cut apart the word cards and lodge sheets.
Two-Key Rite (MFA)	5c	53, 55, 57	1 set. Cut apart the practice QR cards (one per lodge).
Shadow Watch	5c	54, 59, 61, 63	1 set. Read the clue cards; cut apart the lodge answer sheets.
Sealed Scroll Match-Up	6a	60, 65	1 set. Cut apart the cards (one set per lodge).
Trial: The Forged Seal	6c	67-69	1 set. Cut apart the case files (one per lodge). Keep the answer key. (The Warden worksheet is in the per-Warden pack.)
The Grand Tournament	8b	71-73	1 set. Print the lodge slip once per lodge from page 2. Keep the scoring sheet.
Enchanted Objects	7	75	1 copy for you (guide and idea bank). (The Warden worksheet is in the per-Warden pack.)

Total: 76 pages (38 sheets). Spreadsheets (the Cup of the Silver Padlock scoreboard and the Warden Sign-off Tracker) are used on screen and are not in this pack.

Instructor key: Line of Oaths

Suggested discussion points only. Wardens may reasonably disagree, and the reasoning matters more than the spot. Ask one Warden on each side to defend, then ask a Warden in the middle what would tip them.

Scenario	Suggested spot and why	Scout Law ties	Coaching notes
1. Unlocked laptop	Ethical to lock it or tell the owner. Unethical to look around or prank.	Trustworthy, Helpful, Courteous	Leave it alone. Lock it, or tell the owner or an adult.
2. A password you can see	Ethical to look away and tell them to change it. Unethical to use it.	Trustworthy, Loyal, Kind	Never use a password you were not given, even if you remember it.
3. Not-secure website	Ethical to leave and tell an adult or the site owner. Unethical to experiment with it.	Trustworthy, Obedient, Helpful	Do not enter personal information. Watch for Wardens who say "it is fine, it is just a name".
4. A free-money bug	Report it to the game company. Unethical to exploit it, even if "everyone would". This is how responsible reporting works.	Trustworthy, Brave, Helpful	Some Wardens will say it depends on whether the game is competitive. Ask how it affects other players.
5. Mystery USB drive	Ethical to hand it in to a lost-and-found or an adult. Plugging it in is a security risk and snooping.	Trustworthy, Helpful, Clean	Introduce the idea that malware can hide on USB drives. Do not plug it in.
6. Embarrassing photo	Unethical to share or laugh along. Ethical to not forward it, support the person and tell a trusted adult.	Kind, Courteous, Brave	Ask what "Kind" looks like when everyone else is laughing.
7. The sibling prank	Unethical. Logging into another person's account without permission is wrong even as a joke, and may break the law.	Trustworthy, Obedient, Loyal	Ask what could go wrong: lost account, hurt feelings, trouble with parents.
8. Neighbor's open Wi-Fi	It depends. Using someone's network without permission is risky. Ask first, or use a safer option.	Trustworthy, Obedient, Courteous	Great for the IT DEPENDS middle of the line. Let Wardens argue both sides.

- **Setup:** tape a 12-foot line on the floor. Put ETHICAL at one end and UNETHICAL at the other.
- **Pace:** about 1 minute per card. The class plan reads three cards in 4 minutes. Use the extras if time allows or as a bonus round.
- **Points:** +1 per Warden who gives a clear reason, +1 teamwork if a lodge stays together and explains why.
- **Req 2b:** Wardens discuss ethical and unethical online behavior, including an unlocked device, a seen password and an insecure website, and explain why judgment matters.

ETHICAL

Tape to the wall at one end of the line

UNETHICAL

Tape to the wall at the other end of the line

IT DEPENDS

Optional: tape to the middle of the line

Line of Oaths scenario cards

Req 2b. Print on card stock and cut along the dashed lines. Read one card aloud at a time, then Wardens stand on the floor line where they think it belongs.

1 Unlocked laptop

A classmate leaves a laptop open in the library and steps away. It is logged into their game account. What do you do?

Stand on the line: Ethical ← It depends → Unethical

2 A password you can see

Your friend types a password while you sit next to them. You can see every letter. What do you do?

Stand on the line: Ethical ← It depends → Unethical

3 Not-secure website

You find a website with a warning that says it is not secure, and it asks for your name and address. What do you do?

Stand on the line: Ethical ← It depends → Unethical

4 A free-money bug

You discover a glitch in your favorite game that lets you give yourself unlimited coins. Everyone else is playing fair. What do you do?

Stand on the line: Ethical ← It depends → Unethical

5 Mystery USB drive

You find a USB drive in the parking lot with a label that says "Photos". Do you plug it into your computer to see whose it is?

Stand on the line: Ethical ← It depends → Unethical

6 Embarrassing photo

A classmate posts an embarrassing photo of someone else in a group chat. People are laughing and sharing it. What do you do?

Stand on the line: Ethical ← It depends → Unethical

7 The sibling prank

Your friend wants to log into their sister's account to change her profile picture "as a joke" and asks you to help guess her password. What do you do?

Stand on the line: Ethical ← It depends → Unethical

8 Neighbor's open Wi-Fi

Your neighbor's Wi-Fi has no password. Your own internet is down and you want to finish your homework. Do you connect?

Stand on the line: Ethical ← It depends → Unethical

Hunt for Hidden Doors

Instructor guide · Req 1b · 5 minutes in the Digital safety block

The goal. Wardens show they can explain how to protect their digital footprint on social media, apps and games, and can find privacy settings for personal information, photos and location on a real device.

How to run it

1. **Before class:** open each setting on your own phone so you know the exact wording. Menu names change when phones update.
2. **Set the rules (30 seconds):** hold up your device. Look and show, but change nothing you are not comfortable with, and never show a screen with private messages.
3. **Call out a setting.** Read the call-out from the list on page 2.
4. **Lodges race** to find it on a phone and show you. Wardens without a device pair with a teammate.
5. **Explain it (about 30 seconds):** say what the setting protects: your location, your photos or your identity. Then ask the class question.
6. Do the first five in class (about 5 minutes). Use 6 and 7 as a bonus round if time allows.

Points (Cup of the Silver Padlock)

- **+2** for the first lodge to find and show the setting.
- **+1** for any other lodge that finds it within 30 seconds.
- Use the tally sheet on page 3, then enter the points in the Scoreboard workbook.

Tips. Phones differ (iPhone and Android). Let Wardens type the word into the Settings search bar, such as “location”, and they will find it quickly. Settings inside social apps and games vary, so accept any screen that shows who can see posts, tag you or message you. Wardens who do not bring a device can pair with someone, so no one is left out.

Menu paths are written from general knowledge of current iPhone and Android software and may differ by version or brand. Test them on your own device first.

Three Pillars trial: scenario list

Instructor key · Req 3b · 4 minutes in the Fundamentals block

Setup: hang the C, I and A signs in three corners and stand in the fourth. **Play:** read one scenario aloud. Wardens run to the corner they think was hit, then explain. **Points:** +1 for each Warden in the right corner, tallied by lodge, and +1 teamwork for lodges that debate quietly. Six scenarios fit in 4 minutes. Use the rest as spares. **Trick round (rows 13-15):** tell Wardens that some attacks hit more than one corner. If they think two were hit, they stand halfway between those two corners. If they think all three were hit, they stand in the middle. Save the finale (row 15) for the end. Remind Wardens to walk fast.

#	Read aloud	Corner	Why
1	A hacker reads your private messages.	C	The wrong person can see them.
2	A hacker changes your grade in the school's online system.	I	The information was changed.
3	A game server crashes in the middle of a tournament.	A	Players cannot use it when they need it.
4	Someone posts your home address online.	C	Private information was exposed.
5	A virus scrambles your file so you cannot open it.	A	You cannot use your file. (Some Wardens may say I, since the file was altered. Accept it with a good reason.)
6	Someone edits a photo and says it is the original.	I	The photo was changed and passed off as real.
7	A shopping site goes down on the biggest sale day of the year.	A	Customers cannot use it.
8	A stranger finds your password list on a sticky note.	C	Someone who should not see it did.
9	Someone secretly changes your fantasy football lineup.	I	The information was changed without permission.
10	A school's website is flooded with fake traffic and nobody can log in.	A	Real users are locked out.
11	A student secretly copies the test answers from the teacher's computer.	C	Information was seen by someone who should not see it.
12	A glitch changes your saved game score to zero.	I	The data was changed and is no longer correct.
13	TRICK: Attackers lock a hospital's files with ransomware and also copy patient records to sell.	C + A	Files were locked (availability) and private records were stolen (confidentiality).
14	TRICK: A hacker breaks into the school grading system, reads every student's grades, and changes some of them.	C + I	Grades were seen by the wrong person (confidentiality) and changed (integrity).
15	FINALE: Attackers break into a store's server, copy its customer list, change some orders, and then crash the website.	C + I + A	All three were hit: records stolen (C), orders changed (I), and the site down (A). Everyone stands in the middle.

C = Confidentiality, I = Integrity, A = Availability. Some real attacks hit more than one, such as ransomware that locks files and also steals them. Accept a different answer when the Warden gives a good reason.

Call-out list

Read the call-out aloud. Rows 1 to 5 are the main round, and 6 and 7 are bonus.

#	Call out	What it protects	iPhone	Android (menus vary by brand)	Ask the class
1	**Camera location** *“Does your camera app know where you are when you take a photo?”*	Photos can reveal your home, school or favorite places.	Settings > Privacy & Security > Location Services > Camera	Settings > Location > App location permissions > Camera (or Settings > Privacy > Permission manager > Location)	Who could learn where you live from a photo you post?
2	**Location sharing** *“Show me which apps can see your location right now.”*	Your location is how strangers find where you are.	Settings > Privacy & Security > Location Services	Settings > Location > App location permissions	Which apps really need your location? Which do not?
3	**Microphone access** *“Show me which apps can listen through your microphone.”*	Apps that listen when you did not expect it can record private talk.	Settings > Privacy & Security > Microphone	Settings > Privacy > Permission manager > Microphone	Should a flashlight app ever need the microphone?
4	**Who sees my posts** *“In your favorite social app, show me who can see your posts or story.”*	Public posts can be seen and saved by anyone.	In the app: Settings or Settings and privacy > Account privacy, or Story and Post controls	Same: it is set inside each app	What is the difference between friends only and public?
5	**Photo tags** *“Show me the setting that controls who can tag you in photos.”*	Tags tie your name to photos you did not post.	In the app: Settings > Privacy or Profile and tagging	Same: it is set inside each app	What if a friend tags you in something embarrassing?
6	**Who can message me** *“In your favorite game or app, show me who can message or friend you.”*	Strangers can use chat to pressure or trick you.	In the game: Settings > Privacy, Social or Communication	Same: it is set inside each game	Who should be able to message you?
7	**Photo access** *“Show me which apps can see your photo library.”*	Photos hold faces, places and private moments.	Settings > Privacy & Security > Photos	Settings > Privacy > Permission manager > Photos and videos	Is it better to share all photos or only selected ones?

Tally sheet

Write 2, 1 or 0 in the box for each lodge after each call-out. Write lodge names in the header.

#	Setting	Lodge 1	Lodge 2	Lodge 3	Lodge 4	Lodge 5	Lodge 6
1	Camera location						
2	Location sharing						
3	Microphone access						
4	Who sees my posts						
5	Photo tags						
6	Who can message me						
7	Photo access						
	Total						

C

CONFIDENTIALITY

The Veil: only the right eyes may see it · Spell: Velo!

Hang in one corner of the room



INTEGRITY

The Seal: no one has tampered with it · Spell: Sigillum!

Hang in a second corner



AVAILABILITY

The Lantern: it shines whenever you need it · Spell: Lumen!

Hang in a third corner

Game: Locked Vault

Instructor key · Req 4b · 3 minutes in the Threats block

Before class (about 10 minutes)

Print this packet. Cut out the 6 ransom notes (page 2) and the 18 decision cards (pages 3-4). Put the lodge's "files" inside each pouch, zip bag or small box (a treat or a small prize works well) and close it with a combination lock, zip tie or tape. Tape one ransom note to the outside of each. Keep the combination or scissors yourself. Clip each lodge's 3 cards (PAY, RESTORE FROM BACKUP, ASK FOR HELP) together with a paper clip. The packet makes 6 sets for 25 Wardens; print it twice if you run more lodges.

Run the game

1) Hand each lodge a locked pouch and its card set. 2) Say: "Your lodge just got hit by ransomware. The pouch holds your files. Decide what to do. You have 90 seconds." 3) The pouch stays locked while they decide; the note is taped to the outside. 4) At 90 seconds every lodge holds up one card at the same time. 5) Reveal the answers below and award points. Open the pouch for lodges that chose Restore or Ask for Help (the "files" come back). Keep it shut for Pay. Finish with the 3-2-1 mini lesson.

Card	Verdict	Points
RESTORE FROM BACKUP	Best answer when you have a backup. You lose little and the criminals earn nothing.	+2
ASK FOR HELP	Also good. Tell a trusted adult, report it, and let an expert check the device before anything is restored.	+1
PAY	Risky. You may never get your files, you may be hit again, and you pay criminals. Always explain why.	0

3-2-1 backup mini lesson

3 copies of your data · **2** different kinds of storage (for example your computer and an external drive) · **1** copy kept off-site, such as cloud storage or a drive at a relative's house.

Watch out

Use the pretend notes only. Never use a real ransom note, a real wallet address or a real demand. Make it clear to Wardens that the game is pretend, since some are younger and a note can feel scary.

Pairs with the slide "Game: Locked Vault." Wardens who restore or ask for help earn lodge points for the Cup of the Silver Padlock.

Wireshark Demo: Instructor List

Req 4c · 6 minutes in the Threats block · your own laptop on your own network only

Download Wireshark
wireshark.org/#download

Before class

- [ ] Wireshark installed, and you have run one practice capture
- [ ] Wi-Fi interface identified (the one with moving traffic graph)
- [ ] Projector mirrored; font zoomed (View > Zoom In)
- [ ] Pages below tested on the room Wi-Fi; hotspot as backup
- [ ] Close email, chat and other tabs so only demo traffic appears

Test sites

Site	Type	Use it for
`http://neverssl.com`	http only	Main demo: plain-text GET and page content
`http://info.cern.ch`	http only	Backup: first website ever, tiny page, easy to read
`http://httpforever.com`	http only	Backup http page
`http://httpbin.org/forms/post`	http form	Submit fake answers, then show them in the POST (optional)
`http://testphp.vulnweb.com`	practice site	Log in with fake credentials only
`https://www.scouting.org`	https	Contrast: only scrambled bytes

Steps

1. Select the Wi-Fi interface. **Do not start yet.** Say: "We only capture my laptop on my network."
2. Start capture (blue fin). Open `neverssl.com` in a browser.
3. Optional: submit fake answers on `httpbin.org/forms/post`.
4. Stop capture (red square). In the filter bar type `http` and press Enter.
5. Click the GET row. Show the readable request, then right-click > Follow > HTTP Stream to show the page text.
6. For the form: filter `http.request.method == "POST"` and show the typed answers in plain text.
7. Start a new capture (Don't Save). Open the https site. Stop.
8. Filter `tls`. Show the handshake and scrambled data. Nothing readable.
9. Ask: "Why does https help on public Wi-Fi?" Give +1 per good answer.

Useful filters

`http`	plain web traffic	`tls`	encrypted traffic
`http.request.method == "POST"`	form submissions	`ip.addr == x.x.x.x`	one device (your laptop IP)

If it goes wrong

No packets	Wrong interface. Pick the one with the moving graph.
Browser jumps to https	Use `neverssl.com` and type `http://` yourself, or try a private window.
Nothing readable in http	Switch to `info.cern.ch` or the httpbin form.
Wi-Fi blocks the sites	Share your phone hotspot to the laptop.

Say it out loud: capturing other people's traffic is unethical and often illegal. Use fake credentials only, never your real ones.

Ransom notes

Cut apart · one per lodge · tape one to the outside of each locked pouch

!! YOUR FILES ARE LOCKED !!

Your lodge's files have been **encrypted**.

Send **5 PRETEND-COINS** to the Dark Wolf Gang in 24 hours or your files are gone forever.

Do not tell anyone. Do not call for help.

PRACTICE NOTE · Lodge 1 · This is a game. Nothing here is real.

!! YOUR FILES ARE LOCKED !!

Your lodge's files have been **encrypted**.

Send **5 PRETEND-COINS** to the Dark Wolf Gang in 24 hours or your files are gone forever.

Do not tell anyone. Do not call for help.

PRACTICE NOTE · Lodge 2 · This is a game. Nothing here is real.

!! YOUR FILES ARE LOCKED !!

Your lodge's files have been **encrypted**.

Send **5 PRETEND-COINS** to the Dark Wolf Gang in 24 hours or your files are gone forever.

Do not tell anyone. Do not call for help.

PRACTICE NOTE · Lodge 3 · This is a game. Nothing here is real.

!! YOUR FILES ARE LOCKED !!

Your lodge's files have been **encrypted**.

Send **5 PRETEND-COINS** to the Dark Wolf Gang in 24 hours or your files are gone forever.

Do not tell anyone. Do not call for help.

PRACTICE NOTE · Lodge 4 · This is a game. Nothing here is real.

!! YOUR FILES ARE LOCKED !!

Your lodge's files have been **encrypted**.

Send **5 PRETEND-COINS** to the Dark Wolf Gang in 24 hours or your files are gone forever.

Do not tell anyone. Do not call for help.

PRACTICE NOTE · Lodge 5 · This is a game. Nothing here is real.

!! YOUR FILES ARE LOCKED !!

Your lodge's files have been **encrypted**.

Send **5 PRETEND-COINS** to the Dark Wolf Gang in 24 hours or your files are gone forever.

Do not tell anyone. Do not call for help.

PRACTICE NOTE · Lodge 6 · This is a game. Nothing here is real.

Decision cards

Cut apart · 3 cards per lodge · clip each set together

LODGE 1

PAY

Send the money and hope they unlock it.

LODGE 1

RESTORE FROM BACKUP

Wipe the device and bring back a clean copy.

LODGE 1

ASK FOR HELP

Tell a trusted adult and report it.

LODGE 2

PAY

Send the money and hope they unlock it.

LODGE 2

RESTORE FROM BACKUP

Wipe the device and bring back a clean copy.

LODGE 2

ASK FOR HELP

Tell a trusted adult and report it.

LODGE 3

PAY

Send the money and hope they unlock it.

LODGE 3

RESTORE FROM BACKUP

Wipe the device and bring back a clean copy.

LODGE 3

ASK FOR HELP

Tell a trusted adult and report it.

LODGE 4

PAY

Send the money and hope they unlock it.

LODGE 4

RESTORE FROM BACKUP

Wipe the device and bring back a clean copy.

LODGE 4

ASK FOR HELP

Tell a trusted adult and report it.

LODGE 5

PAY

Send the money and hope they unlock it.

LODGE 5

RESTORE FROM BACKUP

Wipe the device and bring back a clean copy.

LODGE 5

ASK FOR HELP

Tell a trusted adult and report it.

LODGE 6

PAY

Send the money and hope they unlock it.

LODGE 6

RESTORE FROM BACKUP

Wipe the device and bring back a clean copy.

LODGE 6

ASK FOR HELP

Tell a trusted adult and report it.

Game: Masked Messengers

Instructor key · Req 4d · 7 minutes in the Threats block

Before class

Print this packet. Cut out the 6 PHISH and 6 LEGIT cards (pages 2 and 3). Each lodge gets one of each, both marked with its lodge number. Cut the 6 message cards (page 4) if you want paper copies, or use the message slides in the deck.

Run it (about 1 minute per message)

1) Show one message and read it aloud. 2) Lodges discuss for 20 seconds. 3) Count 3, 2, 1. 4) Every lodge reveals PHISH or LEGIT at the same time. Ask a lodge that got it right to name the clue, then ask a lodge that got it wrong what made them unsure. **Points:** +1 per lodge with the right call, +1 more for naming the clue. Watch for copying: lodges hold the card to their chest until the count.

#	Message	Answer	Clues to bring out
1	**From: support@netfIix-billing.com** Your account is suspended! Verify your payment within 24 hours or it will be deleted. [Update now]	**PHISH**	The sender spells Netflix with a capital I instead of an l. The message is urgent and scary, and the link is not the real site.
2	**From: troop-leaders@yourtroop.org** Reminder: our meeting moved to Thursday at 7 pm. Bring your lodge box and a flashlight.	**LEGIT**	It comes from a known troop address, has no link asking for secrets, and matches a real meeting change. If unsure, check with a parent or leader.
3	**Text from unknown number** Package delivery failed. Pay a $1.99 fee to reschedule: bit.ly/xk82pa	**PHISH**	A fake delivery text with a small fee and a shortened link. Real carriers do not ask for card details by text.
4	**In-game chat** You won 5,000 V-Bucks! Enter your username and password here to claim your prize.	**PHISH**	Free game currency that asks for a password is a scam. Real rewards never need your password.
5	**From: no-reply@accounts.google.com** New sign-in on your account from a Pixel phone in Florida. If this was you, no action is needed.	**LEGIT**	Legit, with a caution. A real security alert tells you to review activity. The safe habit is to open the app or type the website yourself instead of clicking a link in the message. This one is meant to spark debate.
6	**Text from a friend** omg is this u??? tinyurl.com/pic99x	**PHISH**	A friend's account may have been hacked. The odd urgency, vague 'is this u', and shortened link are warning signs. Message your friend another way to check.

Five red flags to listen for

Strange sender · Urgent or scary · Odd link · Asks for secrets (passwords, codes, gift cards) · Too good to be true. Safe habit: do not click, do not reply, tell an adult, report and delete, and go to the real site by typing the address yourself.

Message 5 is meant to spark debate. Praise Wardens who say "I would go to the real site myself." These are made-up messages for practice.

Game: Count the Open Gates

Instructor key · Req 4f · 2 minutes in the Threats block (a quick call-out)

What it is

Your attack surface is every way someone could reach your information or devices. Each lodge marks the squares its members all have in common, and calls BINGO for three in a row.

Before class

Print pages 2-4 (six lodge cards, each with a different layout) and cut each page in half. Print page 5 for each Warden, because each Warden also writes a personal attack surface list for the Req 4f sign-off. Bring pencils or dry-erase markers.

Run it

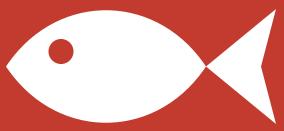
1) Hand out one card per lodge and one worksheet per Warden. 2) Lodges talk it out and mark only the squares every member has. 3) When a lodge calls BINGO, check it and ask them to name one risk for each of the three squares. 4) Each Warden fills in the worksheet. **Points:** +2 for the first bingo, +1 for each additional bingo.

Phone	Lost or stolen; apps that spy or carry malware
Game console	Weak account password; scam messages in chat
Smart speaker	Always listening; default settings left on
School login	Reused or weak password; phishing for your login
Email	Phishing links; account takeover
Shared Wi-Fi	Snooping on traffic; fake hotspots
USB drive	Malware on it; easy to lose
Social media	Oversharing; fake accounts and scams
Smart TV	Old software; camera and mic; default password

Keep it to types of devices and accounts. Wardens should never write down passwords or private details.

PHISH cards

Cut apart · one per lodge



PHISH

Fish: this is a scam

LODGE 1



PHISH

Fish: this is a scam

LODGE 2



PHISH

Fish: this is a scam

LODGE 3



PHISH

Fish: this is a scam

LODGE 4



PHISH

Fish: this is a scam

LODGE 5



PHISH

Fish: this is a scam

LODGE 6

LEGIT cards

Cut apart · one per lodge



LEGIT

Checkmark: this is real

LODGE 1



LEGIT

Checkmark: this is real

LODGE 2



LEGIT

Checkmark: this is real

LODGE 3



LEGIT

Checkmark: this is real

LODGE 4



LEGIT

Checkmark: this is real

LODGE 5



LEGIT

Checkmark: this is real

LODGE 6

Message cards

Optional paper copies · show one at a time

From: support@netfIix-billing.com 1

Your account is suspended! Verify your payment within 24 hours or it will be deleted.

[Update now]

From: troop-leaders@yourtroop.org 2

Reminder: our meeting moved to Thursday at 7 pm. Bring your lodge box and a flashlight.

Text from unknown number 3

Package delivery failed. Pay a $1.99 fee to reschedule: bit.ly/xk82pa

In-game chat 4

You won 5,000 V-Bucks! Enter your username and password here to claim your prize.

From: no-reply@accounts.google.com 5

New sign-in on your account from a Pixel phone in Florida. If this was you, no action is needed.

Text from a friend 6

omg is this u??? tinyurl.com/pic99x

COUNT THE OPEN GATES

LODGE 1

 **Shared Wi-Fi**	 **Email**	 **Smart TV**
 **Social media**	 **School login**	 **Smart speaker**
 **Game console**	 **Phone**	 **USB drive**

Mark every square your whole lodge has in common. Three in a row, call BINGO!

COUNT THE OPEN GATES

LODGE 2

 **Shared Wi-Fi**	 **School login**	 **Smart speaker**
 **Game console**	 **Social media**	 **Smart TV**
 **Phone**	 **USB drive**	 **Email**

Mark every square your whole lodge has in common. Three in a row, call BINGO!

COUNT THE OPEN GATES

LODGE 3

🎮 **Game console**	💬 **Social media**	📺 **Smart TV**
📱 **Phone**	💾 **USB drive**	🏫 **School login**
✉ **Email**	📶 **Shared Wi-Fi**	🔊 **Smart speaker**

Mark every square your whole lodge has in common. Three in a row, call BINGO!

COUNT THE OPEN GATES

LODGE 4

✉ **Email**	💬 **Social media**	📱 **Phone**
🏫 **School login**	📺 **Smart TV**	🎮 **Game console**
🔊 **Smart speaker**	💾 **USB drive**	📶 **Shared Wi-Fi**

Mark every square your whole lodge has in common. Three in a row, call BINGO!

COUNT THE OPEN GATES

LODGE 5

📺 **Smart TV**	🏫 **School login**	🔊 **Smart speaker**
📱 **Phone**	💬 **Social media**	📶 **Shared Wi-Fi**
✉ **Email**	🎮 **Game console**	💾 **USB drive**

Mark every square your whole lodge has in common. Three in a row, call BINGO!

COUNT THE OPEN GATES

LODGE 6

📱 **Phone**	🔊 **Smart speaker**	📺 **Smart TV**
📶 **Shared Wi-Fi**	✉ **Email**	💬 **Social media**
💾 **USB drive**	🏫 **School login**	🎮 **Game console**

Mark every square your whole lodge has in common. Three in a row, call BINGO!

Game: Herald Dispatch

Instructor guide · Req 4e · 5 minutes (2 read, 2 report, 1 vote)

Before class

The six articles (pages 2-7) are in the full Herald Dispatch file in Final Docs, not in the print pack: share each lodge its article on a device, or print them only if you want paper copies. The Warden report sheet (page 8) is in the Per-Warden Print Pack. Page 9 (the answer key) is also in that full file. The articles are rewritten in plain language from real news reports, so they are not graphic. Confirm the facts once more before class, because breach numbers keep being updated.

Run it

1) Hand each lodge its article. Read quietly for 2 minutes (use the timer on the slide). 2) Each Warden gives a 30-second news report to their lodge: what happened, how it happened, what it cost the victim (about 15 seconds each in a lodge of 4-5). Wardens fill in the report sheet for Req 4e. 3) Each lodge votes for its clearest reporter, who gives the report to the whole class. **Points:** +2 for each lodge's winning reporter, +1 for any lodge that identifies the cause correctly.

Tie it back

After the reports, ask: which attacks did we play about today? Phishing (PowerSchool, Discord), password reuse (23andMe), ransomware (M&S, Change Healthcare), and attacks that stop things working (JLR, M&S). Ask which CIA corner each one hit (all hit confidentiality or availability).

Copy-paste prompt: make fresh articles right before class

Paste this into an AI assistant that can search the web (replace the date in brackets). Always check the facts and source links yourself before printing.

```
Act as a news writer for a Scouting America Cybersecurity merit badge class. Today's date is [TODAY'S DATE]. Use web search to find 6 real cyber incidents or data breaches reported in the last 12 months. Pick a mix: one involving a school or students, one involving a gaming or chat app, one caused by password reuse or a stolen login, one ransomware attack that stopped a business, one involving a health or hospital company, and one involving a third-party vendor. For each one, write a one-page news article for Wardens aged 11 to 17: a headline, then 3 short paragraphs (about 150 to 200 words total) in plain language, and 3 vocabulary words with one-sentence definitions. The article must make clear what happened, how it happened, and what it cost the victim. Use only facts that reputable news sources report, with exact dates and numbers. If the cause or the person responsible is unconfirmed or disputed, say so plainly. No graphic detail, no real victims' names, and no instructions for attacking anything. After the six articles, add an answer key table with columns: what happened, how it happened, cost to the victim, and what could have stopped it. Finish with a list of the source URLs you used for each article. Format the result so I can print one article per page.
```

If time is short

Cut the report to 20 seconds, or have only one volunteer per lodge report.

Stories: 1 PowerSchool (schools) · 2 Discord support vendor (gaming and chat) · 3 23andMe (password reuse) · 4 Marks & Spencer (ransomware) · 5 Change Healthcare (health care) · 6 Jaguar Land Rover (factories). Sources are listed on the answer key.

Game: Passphrase Forge

Instructor key · Req 5c (strong password) · 5 minutes in the Defenses block

Before class

Print page 2 and cut it into 48 word cards, or print it twice for extra variety. Print pages 3-4 once. Pencils and paper for each lodge. Remind Wardens: **never type or say a real password in this game.**

Run it

1) Each lodge draws 4 cards face down and turns them over. 2) The lodge writes its 4 words as a passphrase on its lodge sheet, for example blue-tent-owl-river. 3) Reveal 'P@ssw0rd!' next to the strong example. Ask why the longer phrase wins even with no symbols. 4) Lodges name a rule of strong passwords, then every lodge shows its phrase for the vote. **Points:** +2 for the longest, most random passphrase. +1 for any lodge that names a rule (long, unique, no personal facts).

STRONG

`blue-tent-owl-river-maple`

Long, random words, easy to remember, hard to guess.

WEAK

`P@ssw0rd!`

Short and very common. Swapping letters for symbols does not hide it.

Rule of strong passwords	What to say
Long	Length beats tricks. Four or more random words is a good start.
Random	Pick words that do not go together or tell a story about you.
Unique	Never reuse a password. One breach would open every account.
No personal facts	No names, pets, birthdays or teams. Those are easy to guess.
Add a second step	Turn on multi-factor authentication where you can (next game).

Watch out and sign-off

Watch for Wardens choosing personal words. For Req 5c, each Warden must show how to set a strong password on a device. Use a practice account in class, or have Wardens do it on a real account at home with a parent's permission. Never collect or view a real password.

Optional: check the strength of a practice phrase

Go to **bitwarden.com/password-strength** (https://bitwarden.com/password-strength/) and type only the practice phrases built from the cards, such as blue-tent-owl-river-maple and P@ssw0rd!. Never type a real password into any website, even a checker.

Word cards

Cut apart · 48 cards · lodges draw 4

tent	river	owl	maple
compass	ember	trail	canoe
lantern	pebble	thunder	cactus
marble	rocket	meadow	falcon
harbor	pillow	copper	glacier
kettle	saddle	willow	anchor
bridge	cedar	drum	feather
garden	helmet	island	jungle
kayak	ladder	mitten	nickel
orchard	puzzle	quilt	raven
summit	tunnel	umbrella	valley
whistle	yogurt	zipper	acorn

Lodge passphrase sheets

Cut apart or hand one to each lodge

LODGE 1

Our 4 words _____

Our passphrase _____

Letters in it _____ **Rules we followed** ☐ long ☐ random ☐ no personal facts ☐ unique

Rule we can name _____

LODGE 2

Our 4 words _____

Our passphrase _____

Letters in it _____ **Rules we followed** ☐ long ☐ random ☐ no personal facts ☐ unique

Rule we can name _____

LODGE 3

Our 4 words _____

Our passphrase _____

Letters in it _____ **Rules we followed** ☐ long ☐ random ☐ no personal facts ☐ unique

Rule we can name _____

LODGE 4

Our 4 words _____

Our passphrase _____

Letters in it _____ **Rules we followed** ☐ long ☐ random ☐ no personal facts ☐ unique

Rule we can name _____

LODGE 5

Our 4 words _____

Our passphrase _____

Letters in it _____ **Rules we followed** ☐ long ☐ random ☐ no personal facts ☐ unique

Rule we can name _____

LODGE 6

Our 4 words _____

Our passphrase _____

Letters in it _____ **Rules we followed** ☐ long ☐ random ☐ no personal facts ☐ unique

Rule we can name _____

Game: Two-Key Rite (MFA)

Instructor key · Req 5c (multi-factor authentication) · 9 minutes in the Defenses block

What it teaches

Multi-factor authentication adds something you **have** (your phone) to something you **know** (your password). Even if a thief steals your password, they still need the code. Codes are 6 digits and change every 30 seconds.

Before class

Print pages 2-3 (six lodge QR cards plus one spare) and cut them apart. Open **Two-Key Rite - Code Display.html** on your laptop in a browser. It shows every lodge's current code live, so you can check each Warden's phone. Test one scan on your own phone first. The codes are practice-only and are not connected to any real account.

Run it

1) Wardens install an authenticator app from the official store (Microsoft Authenticator or Google Authenticator). Allow about 3 minutes (ask families to install at home first). 2) Each Warden scans their lodge's QR code. 3) They read the 6 digits and show you before they change (30 seconds). Compare to the tile for their lodge on your laptop. 4) The first lodge with every Warden verified wins. **Points:** +2 for the first fully verified lodge, +1 for the second. Wardens without a phone pair up with a teammate.

Watch out

Use the official app stores only. Do not use real accounts. If codes do not match, the phone clock is probably wrong: turn on automatic date and time. Wardens can delete the practice entry from the app afterward. For the Req 5c sign-off, each Warden must show a working MFA code.

Card	Secret (only if a scan fails: add it by hand as a time-based key)
Lodge 1	`3YOM4E2XJIDV4YEW`
Lodge 2	`F6CDGCQ26N7CBUOY`
Lodge 3	`5RUOI55DXSOHUKUT`
Lodge 4	`ZXRZDCHFJYDTWYOL`
Lodge 5	`RYFLAN32XS6OAH5X`
Lodge 6	`O5MPNVMAOPDKK6XN`
Spare	`344DP4PNMBPJVTLU`

Typing the secret by hand: choose Enter a setup key, any name, the secret above, and Time based. The projector page and the cards use the same secrets, so keep this page private.

Game: Shadow Watch

Instructor key · Req 5c (view running processes) · 4 minutes in the Defenses block

What it teaches

Every running program is a **process**. Security people check processes to spot malware. The skill: open the process viewer, sort it, and tell normal from suspicious.

Before class

Print the whole packet. Pages 2 and 3 hold the clue cards (read aloud or hold up) and the lodge answer sheets. Test the steps yourself on your own laptop first. On a phone there is no process viewer, so those Wardens pair with a laptop owner.

Run it

Open the viewer: **Windows** press Ctrl + Shift + Esc (Task Manager). **Mac** press Cmd + Space, type Activity Monitor, press Return. Read the clues one at a time. The first lodge to find each answer raises a hand and shows you. After each clue, discuss what is normal. **Points:** +1 for each clue answered first. Never let Wardens end a process they do not understand.

#	Clue	How to find it	Say
1	**Biggest memory** Find the program using the most memory right now.	Windows: Task Manager > Processes > click the **Memory** column header to sort. Mac: Activity Monitor > **Memory** tab > click the **Memory** column.	Usually a web browser or a big app. Ask: is that what you expected?
2	**Your browser** Find the process for your web browser.	Windows: look under Apps for Chrome, Edge or Firefox. Mac: look for Safari or Google Chrome in the list.	Browsers split into many processes, one per tab or helper. That is normal.
3	**A name you do not know** Pick one process you do not recognize and search its name online.	Pick any unfamiliar name. Search for the exact name plus the word process, and read what a trusted site says. Do not open links from ads.	Most unfamiliar names are normal system helpers. Lookup is how security people check, instead of guessing.
4	**How many processes** Find out how many processes are running.	Windows: Task Manager > **Performance** tab > CPU: the Processes number. Mac: Activity Monitor > **CPU** tab: Processes at the bottom.	There are usually hundreds. That is normal. Ask why a computer needs so many.
5	**Safe to close** Find a process you can safely close.	Open a simple app yourself, such as Calculator, then find it in the list. Close it with its own close button, not by ending a system process.	Only close an app you opened yourself with nothing unsaved. End a task only for a frozen app, with a parent or leader.

Normal

Names you can look up, a few big apps using the most memory, hundreds of processes in total, many browser helpers.

Suspicious signs

A random-letter or look-alike name, very high CPU with nothing open, an app you never installed, or a window that appears and asks you to pay or call someone.

Safety: look, but do not close anything you do not understand. Names here are examples of what to discuss, not a list of what is safe.

Practice QR cards

Cut apart · one per lodge

LODGE 1

Two-Key Rite

1. Open your authenticator app
2. Tap + and scan this code
3. Read the 6 digits and show your counselor before they change

Wardens verified: ☐ ☐ ☐ ☐ ☐ ☐

LODGE 2

Two-Key Rite

1. Open your authenticator app
2. Tap + and scan this code
3. Read the 6 digits and show your counselor before they change

Wardens verified: ☐ ☐ ☐ ☐ ☐ ☐

LODGE 3

Two-Key Rite

1. Open your authenticator app
2. Tap + and scan this code
3. Read the 6 digits and show your counselor before they change

Wardens verified: ☐ ☐ ☐ ☐ ☐ ☐

LODGE 4

Two-Key Rite

1. Open your authenticator app
2. Tap + and scan this code
3. Read the 6 digits and show your counselor before they change

Wardens verified: ☐ ☐ ☐ ☐ ☐ ☐

LODGE 5

Two-Key Rite

1. Open your authenticator app
2. Tap + and scan this code
3. Read the 6 digits and show your counselor before they change

Wardens verified: ☐ ☐ ☐ ☐ ☐ ☐

LODGE 6

Two-Key Rite

1. Open your authenticator app
2. Tap + and scan this code
3. Read the 6 digits and show your counselor before they change

Wardens verified: ☐ ☐ ☐ ☐ ☐ ☐

SPARE

Two-Key Rite

1. Open your authenticator app
2. Tap + and scan this code
3. Show your counselor the 6 digits

Wardens verified: ☐ ☐ ☐ ☐

Clue cards

Read aloud or hold up one at a time

Biggest memory

1

Find the program using the most memory right now.

Your browser

2

Find the process for your web browser.

A name you do not know

3

Pick one process you do not recognize and search its name online.

How many processes

4

Find out how many processes are running.

Safe to close

5

Find a process you can safely close.

Encryption Match-Up

Instructor key · Req 6a · 5 minutes in the Encryption block

What it teaches

Encryption turns readable information into scrambled text that only the right key can unlock. Wardens match each use to the reason it matters, then name the type of encryption.

Before class

Print page 2 once per lodge (6 copies, or 7 for a spare) and cut apart. Each lodge gets three piles: USES (green), REASONS (white) and TYPES (yellow). Shuffle each pile. The slide shows four matches. The fifth, Saved passwords, is a spare for fast lodges.

Run it

1) Lodges match each USE to its REASON for 2 minutes, then add the TYPE. 2) Click through the slide. Each click reveals one correct match with its type. Lodges check theirs. 3) Ask why each type matters. **Points:** +1 for each correct pair and type of encryption.

#	Use	Reason	Type	Say
1	**Online banking**	Keeps your account details private	**HTTPS / TLS**	The padlock and https in the address bar scramble what you send to the bank.
2	**Messaging app**	Only sender and receiver can read	**End-to-end encryption**	Only the two phones hold the keys, so not even the app company can read the message.
3	**Home Wi-Fi**	Keeps neighbors off your network	**WPA2 / WPA3**	Use a strong Wi-Fi password and a modern security setting on the router.
4	**Locked phone**	Protects photos if the phone is lost	**Device encryption**	A phone with a passcode scrambles what is stored on it until it is unlocked.
5	**Saved passwords**	Protects your accounts if your device is stolen	**Encrypted password vault**	A password manager keeps passwords scrambled behind one strong master password.

Note: https tells you the connection is encrypted, not that the site is trustworthy. A phishing site can use https too.

Lodge answer sheets

Cut apart · one per lodge

LODGE 1

	Clue	Our answer	Counselor
1	Biggest memory		
2	Your browser		
3	A name you do not know		
4	How many processes		
5	Safe to close		

☐ We did not close anything we did not understand Suspicious sign we could name: _____

LODGE 2

	Clue	Our answer	Counselor
1	Biggest memory		
2	Your browser		
3	A name you do not know		
4	How many processes		
5	Safe to close		

☐ We did not close anything we did not understand Suspicious sign we could name: _____

LODGE 3

	Clue	Our answer	Counselor
1	Biggest memory		
2	Your browser		
3	A name you do not know		
4	How many processes		
5	Safe to close		

☐ We did not close anything we did not understand Suspicious sign we could name: _____

Lodge answer sheets (continued)

Lodges 4-6

LODGE 4

	Clue	Our answer	Counselor
1	Biggest memory		
2	Your browser		
3	A name you do not know		
4	How many processes		
5	Safe to close		

☐ We did not close anything we did not understand Suspicious sign we could name: _____

LODGE 5

	Clue	Our answer	Counselor
1	Biggest memory		
2	Your browser		
3	A name you do not know		
4	How many processes		
5	Safe to close		

☐ We did not close anything we did not understand Suspicious sign we could name: _____

LODGE 6

	Clue	Our answer	Counselor
1	Biggest memory		
2	Your browser		
3	A name you do not know		
4	How many processes		
5	Safe to close		

☐ We did not close anything we did not understand Suspicious sign we could name: _____

Sealed Scroll match-up cards

Cut apart · one set per lodge

USES	REASONS	TYPES
Online banking	Protects your accounts if your device is stolen	HTTPS / TLS
Messaging app	Keeps your account details private	End-to-end encryption
Home Wi-Fi	Protects photos if the phone is lost	WPA2 / WPA3
Locked phone	Only sender and receiver can read	Device encryption
Saved passwords	Keeps neighbors off your network	Encrypted password vault

Trial: The Forged Seal

Instructor guide · Req 6c (use a hashing tool to show file integrity) · 12 minutes (follows the Wax Seal slide)

What Req 6c asks (hashing option)

Use a hashing tool such as SHA or MD5 to create a checksum for a file, document or text. Have a fellow Warden or the counselor change it, recreate the checksum and compare the two to show file integrity checking. Every Warden who wants Req 6c credit must do the seal, change and compare steps themselves.

Press the Seal	Spot the Forgery	Debrief
6 min	4 min	2 min
Req 6c demo	case files	

Before class

1) Open **Wax Seal Lab.html** on your laptop and test it on a Mac and a Windows device. It runs offline in any browser (nothing is installed, nothing leaves the device). 2) Get it onto Wardens' devices: email it to families ahead of class, share by AirDrop or USB, or have lodges share one device. 3) Print the worksheet (one per Warden), the case file page (one per lodge), and keep the answer key.

Run it

Before you start: the Wax Seal slide has already taught that a hash is a fingerprint for data that cannot be reversed. **Press the Seal:** each Warden hashes a short message in Wax Seal Lab tab 2, copies the first 8 characters, has a teammate change one character, then compares. Check the worksheet and initial it for Req 6c. **Spot the Forgery:** each lodge turns over its case file, types each document exactly into Wax Seal Lab tab 3, enters the posted checksum and marks Untouched or Tampered. **Debrief:** ask what a mismatch tells you, and why a download site would post a checksum.

Scoring (Cup of the Silver Padlock)

Press the Seal worksheet complete **+2** per lodge · first lodge to find the tampered document **+3** · each other lodge that finds it **+1** · explaining why hashing is not encryption **+1**.

Teach these points

• A hash is **not** encryption. Encryption can be undone with a key; a hash is one-way. • Websites store a hash of your password, not the password. • A matching checksum means the file is the same; it does not prove the file is safe. • MD5 and SHA-1 are old and can be fooled on purpose; SHA-256 is the modern choice. • A single typo gives a mismatch too: check the character count first.

Without Wax Seal Lab (own tools)

For a real file: Windows PowerShell `Get-FileHash file.txt -Algorithm SHA256` · Mac Terminal `shasum -a 256 file.txt`. Either way, compare the two results for the original and the changed copy.

Spot the Forgery: Answer Key and Scoring

Counselor copy · do not hand out

Lodge	Tampered	Text as printed	Posted (original)	Actual (printed text)
Lodge 1	**Doc 1**	Camp fee is $54 per Warden.	**66addc55**	**de282a77**
Lodge 2	**Doc 3**	Rain location is the fellowship hail.	**2f66042e**	**9dfb101f**
Lodge 3	**Doc 4**	Counselor email is on the sign-in sheets.	**f5bfc973**	**3f0ce4aa**
Lodge 4	**Doc 2**	Never click links from stranger.	**f55aea31**	**71df43d5**
Lodge 5	**Doc 3**	The hike starts at Trail Marker 21.	**f70d9c57**	**3856ceb1**
Lodge 6	**Doc 4**	Check the fire ring is cold before leaving!	**eb0a6a57**	**ab746dcf**

How each was changed

P1: $45 → $54 (digits swapped). P2: "hall" → "hail". P3: "sheet" → "sheets". P4: "strangers" → "stranger". P5: Trail Marker 12 → 21. P6: "." → "!". All look nearly identical, which is the point of a checksum.

Note

Each posted checksum is the first 8 characters of the SHA-256 hash of the original text. A Warden who types a document with a typo may see a mismatch even on an untouched document: have them check the character count and retype it.

Lodge	Seal worksheet +2	First fake found +3	Others found +1	Hash vs encryption +1	Total
1					
2					
3					
4					
5					
6					

Sign-off check (Req 6c)

☐ Each Warden created a checksum ☐ A teammate or counselor changed the text ☐ Each Warden recreated the checksum and compared ☐ Tracker updated (6c)

Spot the Forgery

Case files · print once, cut apart, one per lodge · hand out face down

Case File · Lodge 1

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Camp fee is $54 per Warden.` 27 characters	`66addc55`	☐ Untouched ☐ Tampered
2	`Bring your own water bottle.` 28 characters	`0371cd61`	☐ Untouched ☐ Tampered
3	`Leave at 7:30 AM from the church lot.` 37 characters	`3f18bbd6`	☐ Untouched ☐ Tampered
4	`Lights out at 9:30 PM sharp.` 28 characters	`a2c6abd1`	☐ Untouched ☐ Tampered

Case File · Lodge 2

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Meet at the flagpole at noon.` 29 characters	`66a47eef`	☐ Untouched ☐ Tampered
2	`Pack lunch for Saturday only.` 29 characters	`500ee3b8`	☐ Untouched ☐ Tampered
3	`Rain location is the fellowship hail.` 37 characters	`2f66042e`	☐ Untouched ☐ Tampered
4	`Parents may pick up at 4:00 PM.` 31 characters	`87e1d235`	☐ Untouched ☐ Tampered

Case File · Lodge 3

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Wi-Fi password is posted at the lodge.` 38 characters	`55fefe6f`	☐ Untouched ☐ Tampered
2	`Do not share your password with anyone.` 39 characters	`b79dc130`	☐ Untouched ☐ Tampered
3	`Update your phone before the trip.` 34 characters	`3178b8b0`	☐ Untouched ☐ Tampered
4	`Counselor email is on the sign-in sheets.` 41 characters	`f5bfc973`	☐ Untouched ☐ Tampered

Case File · Lodge 4

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Report any scam text to an adult.` 33 characters	`e1dbd040`	☐ Untouched ☐ Tampered
2	`Never click links from stranger.` 32 characters	`f55aea31`	☐ Untouched ☐ Tampered
3	`Turn on two-step login today.` 29 characters	`a14a5bee`	☐ Untouched ☐ Tampered
4	`Back up your photos each month.` 31 characters	`3e0cbe58`	☐ Untouched ☐ Tampered

Case File · Lodge 5

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Troop meets every Tuesday at 7 PM.` 34 characters	`45efdd8c`	☐ Untouched ☐ Tampered
2	`Dues are due on the first meeting.` 34 characters	`af5056be`	☐ Untouched ☐ Tampered
3	`The hike starts at Trail Marker 21.` 35 characters	`f70d9c57`	☐ Untouched ☐ Tampered
4	`Wear closed-toe shoes for the hike.` 35 characters	`fe57673d`	☐ Untouched ☐ Tampered

Case File · Lodge 6

One document was tampered with. Find it.

	Document (type it exactly)	Posted checksum	Verdict
1	`Safety check: helmets on before riding.` 39 characters	`4c9e6370`	☐ Untouched ☐ Tampered
2	`First aid kit stays with the leader.` 36 characters	`808a8041`	☐ Untouched ☐ Tampered
3	`Our lodge flag is green and gold.` 33 characters	`c76956c3`	☐ Untouched ☐ Tampered
4	`Check the fire ring is cold before leaving!` 43 characters	`eb0a6a57`	☐ Untouched ☐ Tampered

Cyber Threat Defender

Instructor guide · Req 8b competition · 30 minutes (2:14-2:44)

What Req 8b asks

Participate in a cybersecurity competition with members of your troop, school or another counselor-approved group, either one you design or one on an existing platform. Cyber Threat Defender (CTD) is an existing platform from the UTSA Center for Infrastructure Assurance and Security, free for K-12 education, for ages 11 and up.

Test before class (required)

I could confirm that CTD is a free download, that it runs on Windows PC and Mac through the CIAS Launcher, that the digital game plays against computer opponents, and that CTD's own rules apply (below). I could **not** confirm how long a digital round takes, whether the download needs an account, or whether 25 devices can run it at once. Install it on a Mac and a Windows device, play a full round, and time it. Page: cias.utsa.edu/ctd/play-online. Launcher and games page: cias.utsa.edu/k-12/cybersecurity-games. Questions: cias@utsa.edu.

Launch	Play	Wrap
4 min	21 min	5 min

PREREQUISITE: install before class

Every device that will play must have the CIAS Launcher and Cyber Threat Defender installed and tested before class starts. There is no time to install during the 30 minutes. Install on your own devices and any spares, and ask families to install at home (email below).

Before class

1) Get counselor approval for CTD as the Req 8b competition. 2) Install the CIAS Launcher and CTD on your own device and play. 3) Send the family email below at least a week ahead so Wardens install it at home (Mac and Windows). 4) Print page 2 (one lodge slip per lodge) and the scoring sheet. 5) Plan B: lodges share one laptop and Wardens take turns, or play on your devices.

Run it

Launch (4 min): Wardens open CTD and start a game against the computer. **Play (21 min):** walk the room. Ask each Warden to name one defense card they used and the real-world defense it matches (backups, a patch, MFA, a firewall). **Wrap (5 min):** Wardens report their points; lodges add them; ask which defenses mattered most and which attacks surprised them.

Scoring (Cup of the Silver Padlock)

Lodge total = the sum of its Wardens' points at time-up (a game does not have to finish). Cup points: 1st **+5**, 2nd **+3**, 3rd **+1**. Each Warden who explains a defense card in real terms: **+1**. Counselor breaks ties.

Family email (copy and paste)

Subject: Cybersecurity merit badge: install a free game before class

Hello! At our class on [date] we will play Cyber Threat Defender, a free cybersecurity card game from the University of Texas at San Antonio. Please help your Warden install it before class on a laptop (Windows or Mac): go to cias.utsa.edu/ctd/play-online and follow the steps for the CIAS Launcher. Please play one round so you know how it works. Nothing else is needed, and no personal information should be entered. If it will not install, no problem: Wardens will share a device in class. Thank you!

Cyber Threat Defender: Scoring Sheet

Counselor copy · add to the Cup of the Silver Padlock

Lodge	Warden pts (sum)	Rank	Cup pts (+5/+3/+1)	Defense explained (+1 each)	Notes	Cup total	
1							
2							
3							
4							
5							
6							

Req 8b check

☐ Counselor approved CTD before class ☐ Every Warden took part (own device or shared) ☐ Scores recorded and added to the Cup ☐ Sign-off tracker updated (8b)

Wrap-up questions

Which defense helped the most? Which attack caught you off guard? What real-world habit does that match? How is a game like this different from a real attack?

If devices fail

Pair Wardens. Use one laptop per lodge and rotate every 5 minutes. Score the lodge's best game. The competition still counts as long as every Warden participates.

Cyber Threat Defender: Quick Start

One per lodge · summary of CTD's published rules (cias.utsa.edu/ctd/rules)

Goal. Reach **30 points** while having more points than your opponent. Asset cards earn points. The more assets in your network, the more points you earn each turn.

Setup. Place one Asset – System (Desktop Computer) and one Asset – ISP (ISP Connection) face up. Draw seven cards.

A turn. Play up to **three** cards. Event cards can be played at any time and do not count toward the limit. At the end of your turn, discard down to your hand limit and draw two.

Card types. Asset cards build your network and earn points. **Attack** cards remove points or cards from an opponent. **Defense** cards counter attacks. **Event** cards can change a round. Read each card's text for what it can counter.

You lose if you must draw and your draw pile is empty, or if you reach −45 points.

Think like a defender

For each Defense card you play, say what it protects in real life: a backup, a software update, MFA, a firewall, a strong password. Your counselor will ask.

Safety

Use only the game. Do not enter personal information or accounts. Keep the volume low.

Lodge slip — cut and hand to the counselor at 2:44

Lodge: _____ Warden 1: ____ pts Warden 2: ____ pts Warden 3: ____ pts Warden 4: ____ pts Warden 5: ____ pts **Total:** _____

A defense card we used and the real-world defense it matches: _____

Enchanted Objects: Connected Device Draft

Instructor guide · Req 7 (connected devices) · 7 minutes

Goal

Wardens name everyday objects that connect to the internet (the Internet of Things) and, for four of them, explain why the device is useful, what the risk is, and how to protect it.

Round 1	Round 2	Round 3
1 min	4 min	2 min
lodge draft list	fill in four devices	one device per lodge

Before class

Print one worksheet per Warden (page 2) and have a pencil at each table. Set a 60-second timer for Round 1.

Run it

Round 1 (1 min): each lodge lists as many connected devices as it can on the lodge list at the top of the worksheet. The lodge with the most unique devices earns the bonus. **Round 2 (4 min):** every Warden picks four devices and completes the row: device, why it is useful, the risk, how to protect it. Use the example row as a model. **Round 3 (2 min):** each lodge shares one device and the protection it chose. Push for variety so lodges do not repeat.

Scoring (Cup of the Silver Padlock)

+3 for the longest unique list in Round 1. +1 per Warden who completes four devices. Initial each completed worksheet for Req 7.

Watch for

Wardens who only name phones and TVs. Prompt with cars, doorbells, watches, toys, thermostats, routers, light bulbs and game consoles. Accept any protection that is specific (a password alone is fine, but ask what makes it strong).

Idea bank (answers to accept)

Device	Why it is useful	Risk	Protect it
Smart doorbell / camera	See visitors from anywhere	Hacked camera, shared video	Strong password, MFA, updates
Smart speaker	Hands-free help and music	Always listening, recordings stored	Mute button, review privacy settings, delete history
Game console	Play and chat online	Account takeover, scams, strangers	MFA, parental controls, friends-only chat
Smart TV	Streaming apps	Tracks viewing, weak apps	Update, turn off tracking, strong account password
Smart watch / fitness band	Health and notifications	Location and health data exposed	Lock screen, limit app permissions
Smart thermostat	Saves energy, remote control	Home schedule revealed, hijacked	Update firmware, MFA on the account
Car with Wi-Fi / app	Remote unlock, navigation	Remote takeover, location tracking	Update software, secure the app account
Home router	Connects every device	Default password, old firmware	Change password, update, WPA3/WPA2